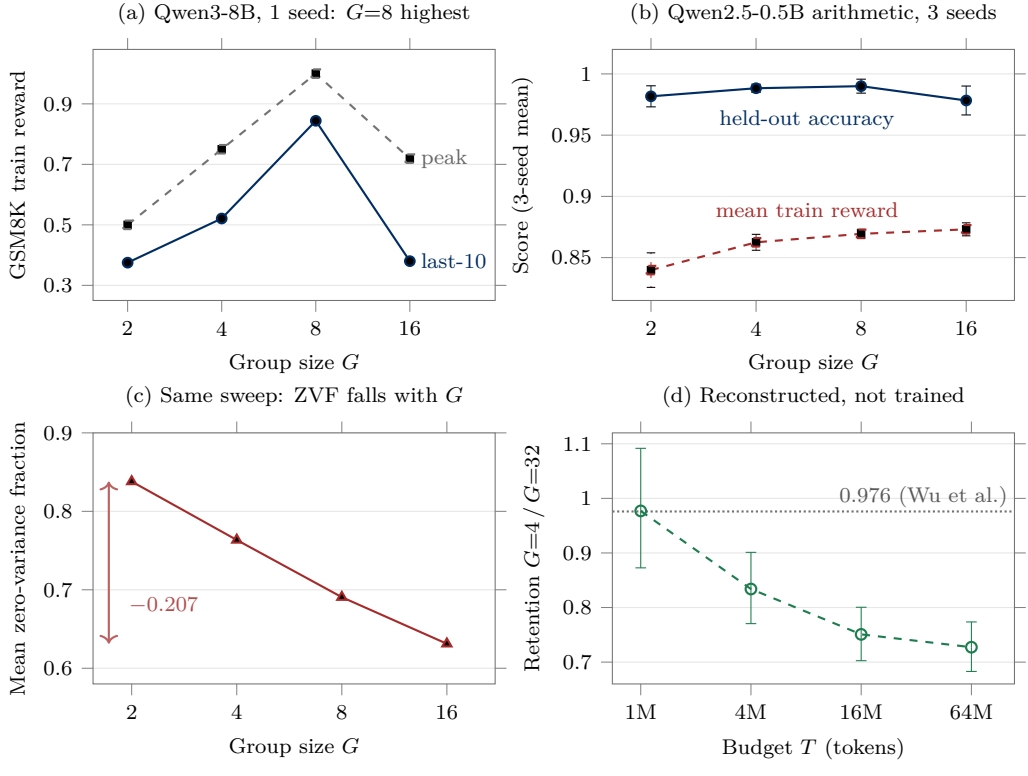
(a) Qwen3-8B, 1 seed: G=8 highest
GSM8K train reward
0.9
0.7
0.5
0.3
peak
last-10
2
4
8
16
Group size G
(b) Qwen2.5-0.5B arithmetic, 3 seeds
Score (3-seed mean)
1
0.95
0.9
0.85
held-out accuracy
mean train reward
2
4
8
16
Group size G
(c) Same sweep: ZVF falls with G
Mean zero-variance fraction
0.9
0.8
0.7
0.6
−0.207
2
4
8
16
Group size G
(d) Reconstructed, not trained
Retention G=4 / G=32
1.1
1
0.9
0.8
0.7
0.976 (Wu et al.)
1M
4M
16M
64M
Budget T (tokens)